

Data: what the ESR maps actually give you, and what's still missing

1. The ESR maps are not "just supply" — they're the best thing in the folder

You asked whether supply is all they give. It isn't. Each of the 37 maps carries **four** things:

On the map	Why it matters
ESR command-area boundary	This is the F1 unlock — which reservoir serves which area
Supply zones (SZ-1 ... SZ-20 per ESR)	Sub-zone granularity <i>inside</i> each command area
Supply hours, colour-coded — 18-24 / 04-08 / 02-04 hours	★ See below
Folder structure = ESR → NMC zone	37 ESRs mapped to 10 NMC zones, free

The supply-hours layer is the find

Some parts of Nagpur get **18 to 24 hours** of water. Others get **2 to 4 hours**. Both appear inside a single ESR's command area — Khamla alone spans all three bands.

Two reasons this matters more than it looks:

- It's the inequality map.** "Which parts of this city get water for 2 hours a day and which get 24" is immediately understandable to any judge, needs no explanation, and as far as I can tell **does not exist anywhere as data** — only as 37 separate PDF pictures.
- It's a contamination risk proxy, with peer-reviewed local backing.** The published Nagpur research on the shift from intermittent to continuous supply found microbial quality improves with 24x7 — because pipes that sit empty between supply hours draw contaminated water in through leaks. So a 2-4 hour zone isn't just inconvenient, it's a *risk indicator*. That links the supply map directly to Q1.

2. What I could and couldn't extract — stated honestly

Extractable automatically, reliable: - ESR name, NMC zone, and the full list of supply-zone codes per ESR - **Blue (04-08 hrs) zones** — the blue tint separates cleanly from everything else

Not reliably extractable: green (18-24) vs yellow (02-04).

The zone tints are semi-transparent washes composited onto satellite imagery — the page is one raster sliced into 115 strips, so there's no separate overlay layer to read. And the basemap is *itself* green over vegetation, which swamps a green tint. I tried two methods (median sampling and bright-surface sampling); both put vegetated zones and green-tinted zones in the same place. I'm not going to hand you a number I can't stand behind.

So: `extract_esr_supply.py` (attached) does the reliable part on all 37 maps locally, renders a preview PNG of each, and writes a CSV where every uncertain row is marked `CHECK`. **Someone opens the previews and fills the green/yellow calls by eye — about a minute per map, so under an hour for all 37.**

That's the right division: the machine does structure, a human does the three-way colour judgement it's actually good at. And the output is a dataset nobody else has.

3. Georeferencing — the honest position

The boundaries are vector, but there's **no georeferencing in the PDFs** — no coordinate grid, no graticule. To place them on a real map you need control points.

Don't try to georeference all 37. For the demo, hand-trace approximate command-area boundaries in geojson.io for the 4–6 ESRs you'll actually show, using the satellite basemap to align. An hour, and it's enough. Label them "approximate, digitised from OCW published maps" — which is both honest and fine.

4. What I fetched for you already

- **347 CGWB groundwater samples** — coordinates, full chemistry incl. SAR and RSC ✓
- **NEERI 2023-24:** 23 river sites, 12 city wells, 10 lakes, 13 STPs — chemistry, coliforms, metals ✓
- **15 raw tables** from the ESR water chapter ✓
- **48-feature GeoJSON** ready to map ✓
- Supply-zone structure for the 8 maps I could stage ✓

5. What I cannot get from here — you'll need to do these

#	What	Why I can't	How long
1	Run <code>extract_esr_supply.py</code> + the eyeball pass	File staging times out on 37 files; and the colour call needs human eyes	~1 hr
2	<code>nagpur.gov.in/disaster-management/</code>	Blocked by robots.txt from this environment	10 min
3	MPCB NRCD STP status page	Same restriction likely	10 min
4	Hand-trace 4-6 ESR boundaries in <code>geojson.io</code>	Needs visual judgement	1 hr
5	<code>r/nagpur harvest</code> — flooding and water complaints with localities	Not reachable from here	30 min
6	NMC ward/zone boundaries as GeoJSON	Need to find the source	20 min
7	Email OCW asking for zone-level quality data	Only you can send it	10 min — and saying you asked is worth points on stage

6. Still to process from what you've already got

- `ESR_Report_22-23.pdf` ☆ — the previous year's NEERI report. Same tables, same parser → **year-on-year trend**. One year is a snapshot; two is a direction. Highest-value item remaining.
- The two academic papers — groundwater near the Nag, and the 2026 surface-water assessment. Independent corroboration plus extra sampling points.
- `MPR_Merged_Oct_2025.pdf` and the unnamed numbered PDFs — not yet identified.
- Lake sampling coordinates (Table 6.4) — chemistry is extracted, locations aren't yet.

7. On the OCWGIS app

Your scan was right and so was your conclusion: it's an authenticated internal ESRI app for utility staff, and no endpoint is exposed. **Leave it there.** Probing an authenticated municipal system is the kind of thing that turns a good project into a problem, and it would gain you nothing you can use on stage.

The legitimate version of the same idea: many Indian ULBs publish **public** ArcGIS REST / ArcGIS Hub open-data endpoints. Searching for a *published* NMC or OCW open-data service is completely fair game, and if one exists it would hand you ward boundaries and possibly the ESR polygons directly. Worth fifteen minutes.